

Tadtapong Chuenchumsaeng

Computer Engineering Student · Applied Computing & Computational Engineering

Bangkok, Thailand · +66 95 325 4203 · 6832083321@student.chula.ac.th

github.com/tadtapongc · linkedin.com/in/tadtapongc · tadtapongc.github.io

Summary

Seeking an internship that applies computing—algorithms, simulation, and software—to other engineering disciplines (e.g., mechanical, aerospace). Computer Engineering undergraduate at Chulalongkorn University; National Olympiad medalist and lead developer of a production CFD automation suite used by a Formula Student team.

Education

Chulalongkorn University

Aug 2025 – Present

B.Eng. in Computer Engineering · GPA 3.53 / 4.00

Mechanical Engineering electives applying computing to physical engineering.

Technical Skills

Programming: Python, C++, MATLAB, Git

Simulation & HPC: OpenFOAM, snappyHexMesh, Ansys Fluent, ParaView, SLURM / MPI, Linux

Design & Engineering: SolidWorks / CAD, carbon-fiber composite fabrication (hand layup, resin infusion, vacuum bagging)

Experience

Rapidamente Formula Student, Chulalongkorn University

May 2026 – Present

Aerodynamics & Composites Engineer

- Co-designed, fabricated, and simulated aerodynamic components for the TSAE2027 race car, bridging composites manufacturing with CFD analysis.
- Fabricated fiberglass negative molds for the front and rear wing main elements, then laid up the parts via wet hand layup with vacuum bagging.
- Selected resin infusion for the wing endplates, matching the process to each part's geometry and target fiber volume.
- Executed steady-state CFD in OpenFOAM (via RapidFOAM) and cross-checked forces in Ansys Fluent at competition airspeed (16.67 m/s) to guide design iterations.

Projects

RapidFOAM / OpenFOAM Case Generator

2026 – Present

Lead Developer & CFD Pipeline Architect · [GitHub](#)

- Built a zero-dependency Python suite automating the full external-aerodynamics CFD workflow (streaming STL ingestion, adaptive meshing, case generation, force post-processing), cutting case setup from 2–3 hours to under 5 seconds.
- Engineered physics-aware domain sizing—two-stage wake refinement, symmetry half-car models with $2\times$ force projection, and moving-ground boundary conditions—powering multi-million-cell HPC runs (SLURM/MPI).
- Implemented automated boundary-layer meshing with near-wall y^+ targeting, a `checkMesh/surfaceCheck` quality gate, and a Three.js Web Studio with live telemetry and remote cluster job management.
- Verified with 384 Python and 62 front-end tests; released open source under the MIT license.

Honors & Activities

Bronze Medalist — 20th Thailand Olympiad in Informatics (TOI20)

2024

SUT Center Representative

Earned a national Bronze Medal in algorithmic programming, progressing from zero experience through POSN Camps 1–2 (C++, dynamic programming, graph theory).

CPxCEDT Openhouse 2025, Chula Engineering

Oct 2025

Workshop Staff & Speaker

Delivered an interactive workshop on number systems and radix conversion to prospective high-school engineering applicants.